

Quartic Causal Renormalization

Checkpoint — 25 August 2026

Abstract

This note records the current rigorous and numerical status of the quartic zero-repair problem. The main structural advance is an exact reduction of two-step causal block matching to an ordinary transport problem, together with an exact renormalization law for the quartic defect. The resulting numerical renormalization data are consistent with a scale-critical regime, but they do *not* prove an $O(\log n)$ upper bound. The load-bearing open step remains a prefix-compatible causal lift with uniformly bounded inventory increment per dyadic scale.

1 Status at a glance

For the explicit quartic pair considered below, the present status is:

$$\boxed{\mathcal{B}_n = \Omega(\log n)} \quad \text{proved,}$$

while

$$\boxed{\mathcal{B}_n = O(\log n)} \quad \text{remains open.}$$

The current renormalization calculations provide no evidence for a runaway superlogarithmic mechanism. Instead, a standardized quartic coordinate appears numerically to enter a narrow scale-critical region. This is evidence only; no implication

$$\text{bounded renormalized defect} \implies \mathcal{B}_{2n} \leq \mathcal{B}_n + C$$

has yet been proved.

2 The explicit quartic core

Define

$$E(z) = \frac{1 + 6z^2 + z^4}{8}, \quad O(z) = \frac{z + z^3}{2}.$$

Thus E and O are the laws of a $\text{Bin}(4, \frac{1}{2})$ random variable conditioned on even and odd parity, respectively. Their difference is

$$\boxed{E(z) - O(z) = \frac{1}{8}(1 - z)^4.}$$

The original pair (P, Q) admits the decomposition

$$G_P(z) = \frac{7}{8}R(z) + \frac{1}{8}E(z), \quad G_Q(z) = \frac{7}{8}R(z) + \frac{1}{8}O(z),$$

where

$$R(z) = \frac{7 + 16z + 10z^2 + 16z^3 + 7z^4}{56}.$$

The scalar two-block identity is

$$E(z)^2 - O(z)^2 = \frac{1}{64}(1 - z^2)^4 = \frac{1}{8}(E(z^2) - O(z^2)).$$

This is a valid static identity, but it does not furnish a causal lift.

3 Exact two-step causal obstruction

Let (X_1, Y_1, X_2, Y_2) be a two-step co-adapted coupling of (E, O) : the first pair is a coupling of E and O , and conditional on the first pair the second pair again has marginals (E, O) .

For a one-step coupling define

$$M(d) := \sup_{\gamma \in \Pi(E, O)} \Pr(X - Y = d).$$

A direct capacity calculation gives

$$M(\pm 1) = \frac{5}{8}, \quad M(\pm 3) = \frac{1}{8}.$$

Equality of two-step totals requires

$$X_2 - Y_2 = Y_1 - X_1.$$

Hence the conditional success probability is at most $5/8$, and this bound is attained.

Theorem 1 (Exact two-step causal block match). *For the quartic parity core,*

$$C_{\text{causal}}^{(2)}(E, O) = \frac{5}{8}.$$

*By contrast, the static maximal common mass of E^{*2} and O^{*2} is*

$$C_{\text{static}}^{(2)}(E, O) = \frac{7}{8}.$$

Therefore

$$C_{\text{static}}^{(2)} - C_{\text{causal}}^{(2)} = \frac{1}{4}.$$

Thus the scalar decomposition

$$\frac{7}{8} \text{ common} + \frac{1}{8} \text{ carry}$$

cannot be realized as a free causal carry mechanism.

For one attaining causal coupling, the block discrepancy

$$D = (Y_1 + Y_2) - (X_1 + X_2)$$

has law

$$\Pr(D = 0) = \frac{5}{8}, \quad \Pr(D = \pm 2) = \frac{1}{8}, \quad \Pr(D = \pm 4) = \frac{1}{16}.$$

4 First residual states

After removing the matched block mass $5/8$, normalizing the residual mass $3/8$, and applying the common lattice rescaling $s \mapsto s/2$, the first renormalized pair is

$$P_1 = \frac{1}{24}(1, 5, 12, 5, 1), \quad Q_1 = \frac{1}{24}(0, 9, 6, 9, 0).$$

Its defect is again quartic:

$$G_{P_1}(z) - G_{Q_1}(z) = \frac{1}{24}(1 - z)^4.$$

The next exact two-step causal block match is

$$C_{\text{causal}}^{(2)}(P_1, Q_1) = \frac{47}{72},$$

so the residual fraction is

$$r_1 = \frac{25}{72}.$$

One exact normalized residual pair is

$$P_2 = \frac{1}{200}(1, 10, 12, 50, 54, 50, 12, 10, 1),$$

$$Q_2 = \frac{1}{200}(0, 0, 44, 28, 56, 28, 44, 0, 0),$$

and

$$G_{P_2}(z) - G_{Q_2}(z) = \frac{(1 - z)^4}{200} (1 + 14z + 18z^2 + 14z^3 + z^4).$$

Thus the pure ray $c(1 - z)^4$ is not invariant, although the fourth-order zero at $z = 1$ is.

For this pair,

$$C_{\text{causal}}^{(2)}(P_2, Q_2) = \frac{14241}{20000}, \quad r_2 = \frac{5759}{20000}.$$

5 Exact reduction of the two-step causal LP

Let $A = (A_x)$ and $B = (B_y)$ be arbitrary finite probability laws on a common integer lattice. For each displacement d define

$$\alpha_d(A, B) := \max_{\gamma \in \Pi(A, B)} \Pr_{\gamma}(X - Y = d).$$

Because the event $X - Y = d$ lies on one transport diagonal,

$$\alpha_d(A, B) = \sum_t \min\{A_t, B_{t-d}\}.$$

Theorem 2 (Transport reduction). *The optimal two-step co-adapted block matching probability is*

$$C_{\text{causal}}^{(2)}(A, B) = \max_{\pi \in \Pi(A, B)} \sum_{x, y} \pi_{xy} \alpha_{y-x}(A, B).$$

Proof. Condition on $(X_1, Y_1) = (x, y)$. Equality of two-step totals is equivalent to

$$X_2 - Y_2 = y - x.$$

The best conditional success probability is therefore $\alpha_{y-x}(A, B)$. Averaging over the first-step coupling π and then optimizing π yields the formula. $\square$

This reduces the causal block problem from an N^4 LP to an ordinary N^2 transport problem. Define

$$c_d(t) := \min\{A_t, B_{t-d}\}.$$

For a chosen first-step transport π , the matched block-total measure is

$$M_\pi(s) = \sum_{x,y} \pi_{xy} c_{y-x}(s - x).$$

If

$$m = M_\pi(1), \quad r = 1 - m,$$

then the normalized residual pair is

$$A' = \frac{A * A - M_\pi}{r}, \quad B' = \frac{B * B - M_\pi}{r}.$$

The renormalization is generally set-valued:

$$(A', B') \in \mathcal{R}(A, B),$$

because distinct optimal transports on the same optimal face can produce different residual states.

6 Quartic defect renormalization

Write

$$D(z) = G_A(z) - G_B(z), \quad S(z) = G_A(z) + G_B(z).$$

Before the common affine lattice normalization,

$$D_{\text{res}}(z) = \frac{D(z)S(z)}{r}.$$

Hence a fourth-order zero at $z = 1$ is preserved.

Define the quartic intensity

$$\eta(A, B) := \frac{1}{4!} \frac{d^4}{dz^4} (G_A(z) - G_B(z)) \Big|_{z=1}.$$

Equivalently,

$$\eta(A, B) = \sum_i \binom{i}{4} (A_i - B_i).$$

Suppose the common affine normalization after residual extraction is

$$s \mapsto \frac{s - c}{g}.$$

Since $G_A(1) = G_B(1) = 1$, the quartic coefficient obeys the exact law

$$\eta_{k+1} = \frac{2}{r_k g_k^4} \eta_k.$$

Equivalently,

$$r_k g_k^4 \eta_{k+1} = 2\eta_k.$$

The first exact values are

$$\eta_0 = \frac{1}{8}, \quad \eta_1 = \frac{1}{24}, \quad \eta_2 = \frac{6}{25}, \quad \eta_3 = \frac{9600}{5759}.$$

7 Scale-free quartic coordinate

Let

$$V_k := \text{Var}(P_k) = \text{Var}(Q_k)$$

and define

$$\chi_k := \frac{\eta_k}{V_k^2}.$$

This quantity is invariant under a common affine rescaling of the lattice.

Let

$$\tilde{V}_{k+1} := g_k^2 V_{k+1}, \quad R_k := \frac{\tilde{V}_{k+1}}{V_k}.$$

Then

$$\frac{\chi_{k+1}}{\chi_k} = \frac{2}{r_k R_k^2}.$$

It is therefore natural to define the critical quantity

$$K_k := r_k R_k^2.$$

The critical hypersurface is exactly

$$K = 2.$$

Indeed,

$$K > 2 \implies \chi_{k+1} < \chi_k, \quad K < 2 \implies \chi_{k+1} > \chi_k,$$

while

$$K = 2 \implies \chi_{k+1} = \chi_k.$$

8 Optimal-face control and the W coordinate

Assume a reflection-symmetric state and replace every optimal transport by its reflected symmetrization. Let μ be the common one-step mean and define the matched central second moment

$$W(\pi) := \sum_s (s - 2\mu)^2 M_\pi(s).$$

Then

$$r \tilde{V}' = 2V - W(\pi).$$

Consequently, on a fixed near-optimal matching face, W is a linear control coordinate for the next residual variance and hence for χ' .

For a prescribed target $\chi_* > 0$, the required value is

$$W_* = 2V - rg^2 \sqrt{\frac{\eta'}{\chi_*}}.$$

If W_* lies between the minimum and maximum values attained on the chosen matching face, a convex combination of the two extreme transports realizes the target exactly.

9 Numerical optimal-face audit

All statements in this section are *numerical*. The transport problems were solved with floating-point HiGHS and checked against the exact renormalization identities.

For the min- W branch, the directly computed standardized quartic coordinate initially follows approximately

$$\chi_k : 0.125 \rightarrow 0.074074 \rightarrow 0.057670 \rightarrow 0.041667 \rightarrow 0.040662 \rightarrow 0.049764 \rightarrow 0.051376.$$

For a max- W branch the raw states differ, but the scale-free coordinate remains of the same order.

At the $N = 33$ state on the target- χ trajectory, the optimal-face extremes give approximately

$$\chi'_{\min} = 0.049839399, \quad \chi'_{\max} = 0.050205208.$$

Hence the target

$$\chi_* = \frac{1}{20} = 0.05$$

is bracketed on this near-optimal face. A convex combination with

$$\lambda \approx 0.440380510$$

produces numerically

$$\chi' = 0.050000000000.$$

At the subsequent $N = 65$ state on this target- χ trajectory,

$$\chi'_{\min} \approx 0.050506786, \quad \chi'_{\max} \approx 0.050711921.$$

Thus $1/20$ is no longer exactly reachable on that particular near-optimal min/max segment; the closest point is the min- W endpoint.

The currently observed behavior is therefore consistent with a narrow scale-critical region near

$$\chi \simeq 0.05,$$

but the value $1/20$ is only a numerical candidate, not a proved fixed point.

10 Numerical validation identities

Each audited transition should satisfy all of the following checks.

First, mass and the first three factorial defects:

$$\sum_i A_i = \sum_i B_i = 1,$$

$$\Delta_j := \sum_i (i)_j (A_i - B_i) \approx 0, \quad j = 0, 1, 2, 3.$$

Second, the quartic renormalization law:

$$\boxed{\eta' = \frac{2\eta}{rg^4}.$$

Third, for a reflection-symmetric matched measure,

$$\boxed{\frac{\tilde{V}'}{V} = \frac{2V - W}{rV}.$$

Finally, the direct and predicted scale-free ratios must agree:

$$\boxed{\frac{\eta'/V'^2}{\eta/V^2} = \frac{2}{r(g^2V'/V)^2}.$$

Agreement of these independent formulas is used as a numerical integrity check; it is not by itself a proof of asymptotic behavior.

11 Numerical implementation checkpoint

The reduced transport formulation is the preferred computational route. The current robust implementation uses:

- active-support compression;
- removal of one redundant marginal equation;
- explicit feasibility checks using the product coupling;
- reflection symmetrization of selected transports;
- actual residual mass rather than the nominal face target;
- direct signed-defect evaluation of η ;
- higher-precision residual algebra where needed;
- explicit verification of all factorial and renormalization identities.

Small negative transport entries produced at floating-point tolerance are treated as solver leakage only when clipping them leaves the marginal constraints within the prescribed numerical tolerance.

No asymptotic claim should be inferred from a solver failure at large support. The transport polytope is mathematically nonempty whenever the input laws are probability measures, because the product coupling $A \otimes B$ is an explicit feasible point.

12 Rigorous lower bound

For a general equal-mean finite-support pair define, for $a > 0$,

$$L(a) = \log \frac{G_P(e^{-a})}{G_Q(e^{-a})},$$

and

$$R(a) = \log \frac{G_P(e^{-2a})}{G_P(e^{-a})^2}.$$

Whenever $L(a) > 0$,

$$c(a) = \frac{L(a)}{a[2L(a) + R(a)]}.$$

The global Laplace barrier gives

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \mathfrak{c}_L(P, Q) := \sup_{a > 0: L(a) > 0} c(a).$$

For the explicit quartic pair, the elementary choice $z = e^{-a} = 1/2$ already yields the rigorous positive constant

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq 0.00766189365248 \dots$$

while numerical scalar optimization gives the larger candidate

$$\mathfrak{c}_L \approx 0.01642637054.$$

Only the former displayed numerical constant is needed to certify $\Omega(\log n)$ without relying on the scalar numerical maximization.

13 What is proved, numerical, and open

Proved

1. The quartic core satisfies

$$E - O = \frac{1}{8}(1 - z)^4.$$

2. The exact two-step causal block match is

$$C_{\text{causal}}^{(2)}(E, O) = \frac{5}{8},$$

whereas the static block match is $7/8$.

3. The two-step co-adapted matching problem reduces exactly to an N^2 transport LP.
4. The quartic defect order is invariant under residual blocking.
5. The exact intensity law is

$$rg^4\eta' = 2\eta.$$

6. The scale-free identity is

$$\chi'/\chi = 2/(rR^2).$$

7. The explicit quartic pair satisfies

$$\mathcal{B}_n = \Omega(\log n).$$

Numerical evidence

1. Several distinct extremal causal histories produce substantially different raw residual shapes but comparable scale-free quartic coordinates.
2. After an initial transient, χ enters a narrow numerical region of order 5×10^{-2} .
3. The value $\chi = 1/20$ is exactly targetable on one audited near-optimal face, but not on the next audited one.
4. No audited trajectory presently exhibits a runaway exponential growth of the standardized quartic defect.

Open

1. Prove a compact invariant region for the full controlled causal renormalization correspondence.
2. Control not only χ but the higher standardized shape and the prefix-dependent phase state.
3. Construct a prefix-compatible positive causal lift across dyadic scales.
4. Prove a bounded doubling increment

$$\boxed{\mathcal{B}_{2n} \leq \mathcal{B}_n + C.}$$

5. Consequently establish, or refute,

$$\boxed{\mathcal{B}_n = \Theta(\log n)}$$

for the explicit quartic pair.

14 Current load-bearing target

The numerical evidence suggests that the scalar quartic coordinate is not the runaway direction. The remaining obstruction is causal phase compatibility.

A suitable theorem would have the following form.

Conjecture 1 (Controlled quartic renormalization). *There exists a compact family $\mathfrak{K}$ of phase-augmented quartic states and a constant $C < \infty$ such that every state $\Theta \in \mathfrak{K}$ admits a two-block causal transition*

$$\Theta' \in \mathcal{R}(\Theta) \cap \mathfrak{K}$$

whose additional zero-repair inventory cost is at most C .

Such a theorem would imply along dyadic scales

$$\mathcal{B}_{2K} \leq CK + O(1),$$

and hence

$$\mathcal{B}_n = O(\log n).$$

Combined with the proved Laplace lower bound, this would give

$$\boxed{\mathcal{B}_n = \Theta(\log n).}$$

At present this implication is the principal unresolved step. The renormalization calculations identify a plausible critical state space, but they do not yet provide the required prefix-compatible inventory lift.